Allen Firth

Senior Software Engineer and Solution Architect | C#/.NET, Azure, TypeScript, React, Angular

Durban, South Africa (remote; UTC+2, full UK and EU working-hours overlap) | +27 60 840 7793 | allen@codestream.co.za
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PROFILE

Senior full-stack engineer and solution architect with 25+ years (since 2001) of building production systems for trading, capital markets, law, insurance, sport and retail clients, including LSEG, Uniper, Anglo American, Clifford Chance, EDF Trading and Chelsea FC. I specialise in legacy-to-web rewrites, real-time data (SignalR, WebSockets, Azure Event Hubs) and event-driven .NET and Azure architectures. I have been a CTO and co-founded a SaaS platform that was acquired in 2022. After six and a half years contracting in London, I now deliver remotely through my consultancy, CodeStream. Since June 2025 I have worked AI-first: coding agents handle scaffolding, refactoring and tests, while I review the architecture, security and every merge.

> **Engagement model:** independent contractor, B2B through my consultancy CodeStream; fully remote.

KEY SKILLS

Backend: C#, .NET / ASP.NET Core, Web API, SignalR, CQRS (MediatR), Entity Framework, Dapper, event sourcing (Marten, Wolverine), NServiceBus, Node.js, GraphQL

Frontend: TypeScript, React, React Native, Angular (NgRx, NGXS), Zustand, Zod, Web Components, WebSockets, Tailwind

Cloud and DevOps: Microsoft Azure (Functions, Event Hubs, Service Bus, Data Factory, Synapse, Azure AD / AD B2C, Application Insights), Azure DevOps, GitHub Actions, CI/CD, Docker

Data: SQL Server / Azure SQL, PostgreSQL, Azure Cosmos DB, MongoDB, Power BI

Architecture and practice: solution architecture, distributed and event-driven systems, multi-tenant SaaS, legacy modernisation, TDD, automated testing, Agile / Scrum / SAFe, technical leadership, stakeholder and requirements analysis

AI-augmented delivery: AI coding agents and agentic workflows, AI-assisted TDD and refactoring, human-in-the-loop review

SELECTED ACHIEVEMENTS

- **Uniper:** led the full-stack rewrite of a failed PPA valuation application, making critical paths 10 times faster.
- **LSEG:** rebuilt TM3 for LSEG Workspace with live trade streaming; multiple releases moved clients off the legacy product.
- **DigitalTwin:** co-founded and architected a multi-tenant IoT SaaS platform, acquired by a partner in 2022.
- **Chelsea FC:** live GPS telemetry on Azure, taking live tests from patchy data to reliable tracking of 20+ players.
- **Stuart Peters:** a 130-page PLM analysis led the business to adopt Lectra Kubix Link, which I recommended.

CONTRACT AND CAREER HISTORY

Lead Front-End Developer | corfinancial (Salerio)

Via Digiterre | London, UK (remote) | Capital markets — **Feb 2026 – Present**

- Leading the rewrite of Salerio, a trade management and exception-handling platform, from a legacy C# Windows UI to a React and TypeScript web application.
- One of two developers; about 90% of screens and functionality delivered so far, AI-assisted with full code review.

STACK React · TypeScript · Zustand · Zod

Business Analyst and Software Architect | Stuart Peters (PLM replacement)

Via CodeStream | London, UK and Hong Kong | Fashion design and sourcing — **Apr 2025 – Dec 2025**

- Interviewed stakeholders on site in London and Hong Kong and wrote a 130-page report on business processes, PLM gaps and the way forward. The business adopted my recommendation, Lectra Kubix Link.
- Took over support and maintenance of the React and MongoDB application that manages supplier workflows.

STACK React · MongoDB

Full-Stack Consultant | corfinancial (bitarisk REG9 compliance module)

Via Digiterre | Remote | Wealth and portfolio management — **Jul 2024 – Mar 2025**

- Built the REG9 compliance module for bitarisk, a private client portfolio management system, leading UI delivery within a fully remote team of five and extending the Angular UI with new components integrated with backend APIs.

STACK Angular · TypeScript

Full-Stack Consultant | London Stock Exchange Group (TM3)

Via Digiterre | Remote | Financial markets, fixed income — **Jun 2023 – Jul 2024**

- Redeveloped TM3 (Municipal Market Monitor), a US municipal bond information product, for LSEG Workspace, integrating multiple data sources and streaming live trades to clients over WebSockets.
- Worked closely with the business; multiple production releases let LSEG move clients off the legacy product.

STACK Web Components · TypeScript · WebSockets · Node.js · GraphQL

Lead Full-Stack Consultant | Uniper (PPA Model Valuation Desktop)

Via Digiterre | London, UK (remote) | Energy trading — **Jul 2022 – May 2023**

- Led a full-stack team rewriting the PPA valuation application after the previous version failed to meet business requirements. Delivered model input handling and timeseries visualisation and analysis for renewable energy deals.
- Rebuilt the critical paths, making key functional areas 10 times faster.

STACK Angular · NgRx · TypeScript · SignalR · C# · ASP.NET Core · Azure Cosmos DB

Full-Stack Developer | Clifford Chance (PartnerHub)

Via CodeStream | London, UK (remote) | Legal, global partner remuneration — **Jan 2022 – Jul 2022**

- Developed PartnerHub, the firm's global partner remuneration system across multiple tax regimes, in a SAFe team with domain experts. Implemented interest calculations, including ad hoc events and quarterly reconciliation.
- Built the Angular proof of concept that started the ASP.NET MVC to Angular and ASP.NET Core API migration, then screens for interest reconciliation, manual adjustments, unit pre-allocation and capital loans, plus RBAC and reusable components.

STACK Angular · TypeScript · C# · ASP.NET Core · ASP.NET MVC · SQL Server · Azure

Full-Stack Consultant | Anglo American (Live Exposure Reporting)

Via Digiterre | London, UK (remote) | Commodities trading — **Jul 2021 – Dec 2021**

- Built intraday exposure reporting for traders with multi-dimensional filtering, pivoting and drill-down from book and strategy to trade. Trades were broadcast to the UI in real time with SignalR, including pause-and-snapshot investigation.

STACK Angular · AG Grid · TypeScript · C# · ASP.NET Core · SignalR · Azure Synapse

Co-founder, Architect and Lead Developer | DigitalTwin

Multi-tenant IoT SaaS for live asset tracking — **2020 – 2022**

- Designed and built a multi-tenant platform tracking Bluetooth Low Energy (BLE) tagged assets in real time through fixed gateways, with live inventory, missing-asset alerts and mobile inventory scans.
- Architected the APIs as CQRS (MediatR) with separate read and write models, audit trails and pre-aggregated reporting; Azure Functions for messaging and PDFs; SignalR for live location updates.
- A partner productised it as a retail trolley management system and acquired it in 2022; I led the full technical handover.

STACK ASP.NET Core 5 · SQL Server · EF / Dapper · Azure Functions · SignalR · Angular (NGXS) · Xamarin · Azure AD B2C · Azure DevOps

Software Developer | Chelsea FC (live player telemetry)

Direct engagement via CodeStream | London, UK (remote) | Sports performance analytics — **Aug 2018 – Aug 2020**

- Built an Azure Function aggregating live 10 Hz GPS player data from Azure Event Hubs into per-player, per-second metrics for the analytics API and live Power BI.
- Built the session loader, diagnostics, CI/CD and Azure AD migration; live tests went from patchy to reliable for 20+ players.

STACK C# · Azure Functions · Azure Event Hubs · SQL Server · Power BI · Azure DevOps · Azure AD

Consultant Developer | The SPAR Group

Via CodeStream | Durban, South Africa | Retail — **2017 – 2021**

- Data lake and reporting: centralised legacy operational data into an Azure SQL data lake with ASP.NET Web API reporting endpoints, moving towards Data Factory and Synapse.
- Store and vendor master data: built distributed master-data applications across SAP, iSeries/AS400 and SQL Server, with Angular UIs and IBM Integration Bus; coordinated sprints across teams.

STACK C# · ASP.NET Web API · Azure SQL · Data Factory · Synapse · Angular · SAP APIs · IBM Integration Bus

CTO, then Technology Partner | SMEasy

Durban, South Africa | Online accounting SaaS for SMEs — **2015 – 2024**

- CTO (2015–2017): advised the board on technology strategy; owned architecture; managed developers and contractors.
- Technology partner (2017–2024): delivered the Absa bank white-label edition, an OAuth partner API for fintech integrations, reporting and exports, and ran Azure, Azure DevOps and SQL Server operations.
- Technical lead (2023) for an outsourced Angular front-end rewrite to go-live in Jan 2024; handed over GitHub, pipelines and Azure.

Other CodeStream Engagements and Products — **2017 – Present**

- **Veridot:** micro-dotting APIs on Azure. **CashRewards** (AU): Angular coupon site. **PROFiltr:** Node.js/MongoDB marketplace.
- **Own products (2024–2026):** ContractorPro (React Native, event-sourced Marten/PostgreSQL), Cosplay Platform, CarwashPro.

Earlier Contracts, London — **2008 – 2015**

- **Via Digiterre (2011–2015):** team lead for E.ON's transfer pricing tool (London, Düsseldorf and India teams); master data and initial-margin reconciliation for EDF Trading; introduced CI/CD, automated testing and agile practices at M&G Investments.
- **MAN Investments (2010–11):** fund accounting, fee and FX integration over pub/sub messaging. **Brit Insurance (2009–10):** policy management on NServiceBus. **Freshfields (2008–09):** event and budgeting tools, ASP.NET MVC.

EDUCATION AND CERTIFICATIONS

Bachelor of Commerce, specialising in Informatics | University of South Africa (UNISA), 2004

MCSD, Microsoft Certified Solution Developer | Microsoft, 2001